Portable Aluminum-Seawater Energy System

Abstract

A portable, off-grid system utilizes waste aluminum pellets activated with a gallium-indium alloy or alternatives such as caffeine or imidazole, reacting with seawater to produce desalinated water, hydrogen fuel, and usable heat or electricity. The exothermic reaction drives a closed-loop distillation and desalination process via a heat exchanger and reverse osmosis membrane, capturing hydrogen for fuel cell use and recycling byproducts for sustainability. The system is designed for remote, disaster relief, military, and maritime applications, providing approximately 1,000 liters of fresh water per day alongside energy outputs, with a business model focused on low-cost units and recurring refills.

Contents

1 Field of the Invention

The present invention relates to portable energy and water purification systems, specifically off-grid devices that harness chemical reactions between activated aluminum and seawater to generate hydrogen, heat, electricity, and desalinated water in a sustainable, closed-loop manner.

2 Background of the Invention

Access to clean water and reliable energy remains a critical challenge in off-grid, remote, or disaster-stricken areas. Traditional desalination methods, such as reverse osmosis (RO), often require significant electrical input from fossil fuels or grid power, contributing to carbon emissions and logistical complexities. Hydrogen production typically relies on electrolysis or steam reforming, which are energy-intensive and not feasible in remote settings without infrastructure.

Waste aluminum, abundant from scrap sources, can react with water to produce hydrogen and heat, but the natural oxide layer on aluminum inhibits this reaction, particularly in seawater. Recent advancements, such as activation with gallium-indium alloys or organic compounds like imidazole from coffee grounds, have enabled rapid reactions in saline environments. However, existing systems focus primarily on hydrogen generation and do not integrate desalination or multi-output utilization in a portable format.

The present invention addresses these gaps by providing a self-contained system that uses waste aluminum and seawater to simultaneously produce fresh water, hydrogen fuel, and energy, reducing reliance on external power and enabling circular byproduct recycling. This innovation targets markets including disaster relief, military operations, offshore rigs, islands, ships, and survival applications, with an estimated market potential of $1.23.8 billion at a 13% CAGR.

3 Summary of the Invention

The invention comprises a portable system where activated waste aluminum pellets react with seawater in an exothermic process, generating hydrogen gas, heat, and aluminum hydroxide. The heat is captured via a heat exchanger to drive a turbine-powered reverse osmosis unit for desalination, producing fresh water and brine. Hydrogen is stored and used in a fuel cell for electricity, enabling self-sustaining operation or external power supply. Activation methods include gallium-indium alloys, tin-bismuth coatings, or caffeine/imidazole solutions for cost-effective, environmentally friendly alternatives.

Key features include:

- Off-grid operation using abundant waste materials.
- Dual outputs: ∼1,000 liters fresh water/day and hydrogen for ∼10 kW fuel cell output.
- Recyclable byproducts and recoverable activators for sustainability.
- Modular design for portability and scalability.

The system overcomes challenges in prior art by integrating heat utilization for desalination, minimizing activation costs through recovery, and ensuring safe byproduct handling.

4 Detailed Description of the Invention

4.1 Core Components and Process

The system is a compact, portable unit (e.g., backpack or cart-sized) consisting of:

- An aluminum pellet reservoir for storing pre-treated waste aluminum (24 mm pellets).
- A reaction chamber where seawater is introduced to initiate the aluminum-water reaction.
- A heat exchanger to capture exothermic heat using a working fluid (e.g., isobutane, ammonia, or steam).
- A turbine or expander connected to a reverse osmosis (RO) membrane system for desalination.
- Hydrogen capture and storage modules (e.g., compression tanks at 350700 bar or liquid storage).
- A fuel cell for converting stored hydrogen to electricity.
- Filtration and settling tanks for byproduct management.

The process operates as follows:

1. Aluminum Activation: Waste aluminum pellets are pre-treated to disrupt the oxide layer. Methods include:
   - Application of gallium-indium (Ga-In) eutectic alloy via wiping or cold spray to form sparse islands for penetration without clumping.
   - Gallium-free alternatives: Tin-bismuth (Sn-Bi) coatings for micro-galvanic activation, or immersion in caffeine/imidazole solutions (e.g., derived from coffee grounds) to accelerate reactions in seawater.
   - Preparation: Pellets are chopped to 24 mm, cleaned with mild NaOH, and activated to achieve high surface area and rapid reaction rates (90100% hydrogen yield in minutes, as demonstrated in related research).
2. Reaction Phase: Activated aluminum reacts with seawater in the chamber:
   - Chemical equation: ${}_2Al + {}_6H_2O \longrightarrow {}_2Al(OH)_3 + {}_3H_2$ + heat.
   - Outputs: Hydrogen gas (captured via gas separator), thermal energy ($\sim$139 kJ/g H_2), and aluminum hydroxide sludge.
   - The reaction proceeds efficiently in saline water without prior desalination, though slightly slower than in pure water.

3. Heat Capture and Desalination: Exothermic heat is transferred to a working fluid in the heat exchanger, driving an expansion turbine to generate mechanical pressure.
    - This pressure powers the RO membrane, forcing seawater through to produce fresh water (permeate) and concentrated brine (reject).
    - Brine can be recycled as reaction feedstock or discharged with environmental safeguards (e.g., diffusion to minimize marine impact).
    - Excess heat can be used for additional processes, such as distillation or space heating.
4. Hydrogen Storage and Utilization: Captured hydrogen is compressed or liquefied for storage and fed into a proton exchange membrane (PEM) fuel cell.
    - Electricity generated powers the system (e.g., pumps, controls) or external loads.
    - Optional: Integration with metal hydrides for safer, lower-pressure storage.
5. Byproduct Management and Sustainability:
    - Aluminum hydroxide is filtered and collected; it can be recycled into alum for water treatment, boehmite for semiconductors, or neutralized for safe disposal.
    - Activators like Ga-In are recoverable (up to 99% efficiency) for reuse, reducing costs.
    - The system emits no carbon and uses waste aluminum, promoting circular economy principles.

4.2 System Specifications

- Capacity: $\sim$1,000 liters fresh water/day; hydrogen sufficient for $\sim$10 kW continuous fuel cell output.
- Dimensions: Portable unit (e.g., 50100 kg, modular for transport).
- Capital Cost: $10,00015,000 per unit.
- Operating Cost: $0.300.50 per 10 liters, primarily from pellet and activator refills.
- Business Model: Low-margin hardware sales; revenue from proprietary pre-treated pellet refills (similar to pod-based systems) and partnerships with scrap dealers for aluminum supply.

4.3 Power and Energy Calculations

The following calculations demonstrate the systems energy balance and scalability. All values are based on standard thermodynamic data and assume a 50% efficient PEM fuel cell.

4.3.1 Hydrogen Requirements for 10 kW Output

Step	Description	Calculation
1	Fuel cell efficiency	Assumed
2	Required H_2 energy input	10 kW / 0.5
3	H_2 energy density (lower heating value)	Standard value
4	H_2 mass needed	20 kW / 33.3 kWh/kg
5	Volume at STP (density 0.08988 kg/m^3)	0.6 kg / 0.08988 kg/m^3
6	Compressed volume at 350 bar ($\sim$0.023 kg/L density)	0.6 kg / 0.023 kg/L
7	Compressed volume at 700 bar ($\sim$0.040 kg/L density)	0.6 kg / 0.040 kg/L
8	Liquid H_2 volume ($\sim$0.070 kg/L density)	0.6 kg / 0.070 kg/L

Table 1: Hydrogen Requirements for 10 kW Output

Results: 50%, 20 kW, 33.3 kWh/kg, $\sim$0.6 kg/hour, $\sim$6,675 liters/hour, $\sim$26 liters/hour, $\sim$15 liters/hour, $\sim$8.6 liters/hour.

4.3.2 Aluminum Requirements for Hydrogen Production

Step	Description	Calculation
1	Reaction stoichiometry	2Al (54 g) produces $3H_2$ (6 g)
2	Aluminum for 0.6 kg H_2	600 g H_2 / 0.111 g H_2/g Al
3	Water consumed (from reaction)	Scaled from stoichiometry ($6H_2O$ per 2Al)

Table 2: Aluminum Requirements for Hydrogen Production

Results: H_2 yield: 0.111 g H_2/g Al, $\sim$5.4 kg Al/hour, $\sim$10.8 liters/hour.

4.3.3 Heat for Desalination

Step	Description	Calculation
1	Reaction heat output	$\sim$139 kJ/g H_2 (based on $\Delta H \approx -418$ kJ/mol A
2	RO energy needs	$\sim$3 kWh/m^3 fresh water
3	Heat-to-power efficiency (e.g., ORC turbine)	1015%

Table 3: Heat for Desalination

Results: $\sim$83,400 kJ for 0.6 kg H_2 ($\sim$23.2 kWh thermal/hour), $\sim$0.126 kWh/hour for 1,000 L/day ($\sim$0.042 m^3/hour), $\sim$2.33.5 kWh available/hour (excess beyond RO needs).

These calculations confirm the systems self-sufficiency, with excess energy for scaling or auxiliary uses.

4.4 Scientific Basis and Novelty

The invention builds on established aluminum activation techniques (e.g., Ga-In penetration or imidazole catalysis) for seawater-compatible hydrogen production, achieving high yields without carbon emissions. Uniqueness lies in the integrated use of reaction heat to drive RO desalination, creating a multi-output system not found in prior art. Challenges such as activator recovery, sludge handling, and efficiency are addressed through recyclable components and optimized designs (e.g., filtration tanks, supercritical CO_2 fluids).

Potential enhancements include solar-assisted startup, safer hydride storage, and brine diffusion for environmental compliance.

5 Diagrams

5.1 System Process Flow Diagram

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Seawater Input --> Reaction Chamber (Activated Al Pellets + Seawater)
                          |
                          v
Exothermic Reaction: H2 Gas + Heat + Al(OH)3 Sludge
                          |
         +----------------+----------------+
         |                                 |
         v                                 v
Hydrogen Capture --> Storage (Compressed/Liquid) --> Fuel Cell --> Electricity Output
         |
Heat Exchanger --> Turbine --> RO Membrane --> Fresh Water Output + Brine (Recycle/Di
                          |
                          v
Byproduct Filtration --> Al(OH)3 Recycling + Activator Recovery
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5.2 Energy Balance Overview (Simplified Chart)

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Input: Waste Al (5.4 kg/hr) + Seawater (10.8 L/hr)
|
v
Reaction Outputs:
- H2: 0.6 kg/hr (for 10 kW electricity via 50% efficient fuel cell)
- Heat: 23.2 kWh thermal/hr (powers RO for 1,000 L/day fresh water + excess 2.3-3.5 k
|
v
Outputs: Fresh Water (1,000 L/day) + Electricity (10 kW) + Recyclable Byproducts
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6 Claims

1. A portable energy and desalination system comprising: a reaction chamber for mixing activated waste aluminum pellets with seawater to produce hydrogen, heat, and aluminum hydroxide via an exothermic reaction; a heat exchanger to capture

said heat and drive a turbine-powered reverse osmosis membrane for producing desalinated water; a hydrogen storage and fuel cell module for generating electricity; wherein the aluminum is activated using gallium-indium alloy, tin-bismuth coating, or caffeine/imidazole solutions.

2. The system of claim 1, wherein the aluminum pellets are pre-treated to 24 mm size and activated with sparse islands of gallium-indium for recoverable, efficient reaction in saline water.

3. The system of claim 1, further comprising byproduct recycling means for converting aluminum hydroxide into alum or boehmite.

4. The system of claim 1, configured to produce at least 1,000 liters of fresh water per day and hydrogen sufficient for 10 kW fuel cell output, operating off-grid with waste aluminum inputs.

5. A method for off-grid water and energy production comprising: activating waste aluminum with a gallium-free agent such as caffeine; reacting said aluminum with seawater to generate hydrogen and heat; utilizing said heat to power reverse osmosis desalination; storing and converting hydrogen to electricity via a fuel cell.

6. The method of claim 5, further including recovering activators and recycling byproducts for sustainability.

7. The system of claim 1, wherein hydrogen storage is achieved via compression at 350700 bar or liquefaction, with volumes calculated based on production rates of approximately 0.6 kg/hour.